

**Identity and Access Management (IAM) Project
Active Directory (On-Premises) Deployment For
GoldCorp Inc**

Implementation of On-Premises Active Directory for Centralized Identity and Access
Management

Organization: GoldCorp

Date: September 21, 2025

Prepared By: Tega T. Gold (Cybersecurity Analyst)

1. Project Overview

This project demonstrates the deployment of **Active Directory Domain Services (AD DS)** in a virtual lab environment for **Goldcorp**. The objective was to establish a centralized **Identity and Access Management (IAM)** system that enforces security policies, manages user access, and applies the **principle of least privilege**.

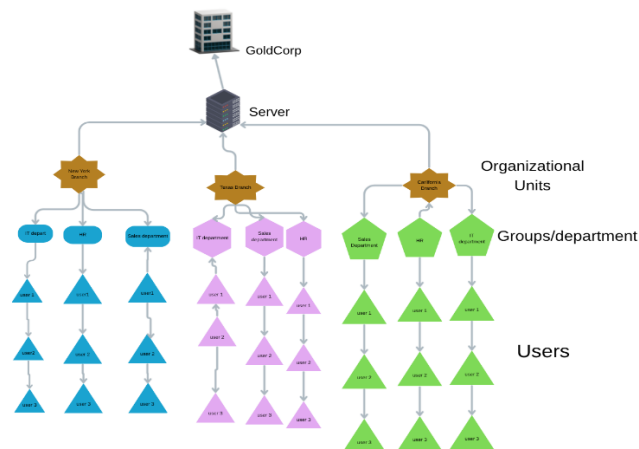
The implementation involved:

- Setting up an on-premises **Domain Controller**.
- Creating **Organizational Units (OUs)** to reflect business branches.
- Defining **security groups** aligned with departments.
- **Provisioning user accounts**.
- Applying **Group Policy Objects (GPOs)** to enforce security and operational restrictions.

2. Company IT Structure

The simulated environment reflected a small IT services firm with distributed offices. Below is a Company Structure and Visual representation:

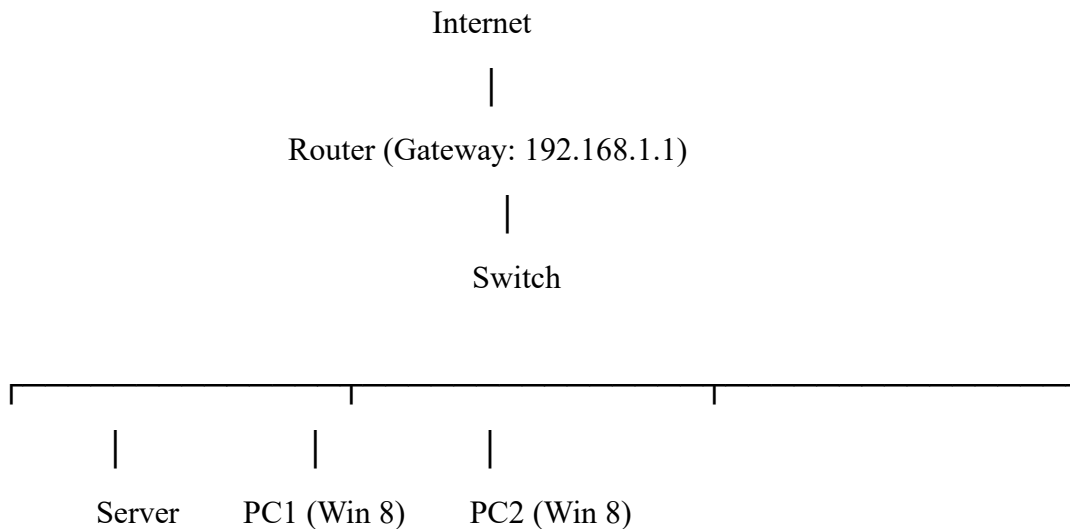
- **1 x Windows Server** – Domain Controller (AD DS + DNS)
- **2 x Client PCs** – Windows 8
- **Three regional OUs (representing branches)** – Texas, California, and New York
- **Departmental Groups** – Created within each OU to represent business functions



3. Project Objectives

- Deploy **Active Directory Domain Services (AD DS)** for centralized IAM.
- Configure **Organizational Units (OUs)** to mirror Goldcorp’s branch structure.
- Create **departmental groups** for access management.
- Provision **security groups** to manage access by department.
- Create **user accounts** and assign them to relevant groups.
- Apply **Group Policies** to enforce access restrictions.
- Demonstrate IAM governance in an on-premises enterprise setup.

4. Network Design



Device	IP Address	Role
Windows Server	192.168.1.66	AD Domain Controller (DC)
Windows 8 PC 1	DHCP	Client (Texas OU – IT)

Windows 8 PC 2	DHCP	Client (New York OU – Sales)
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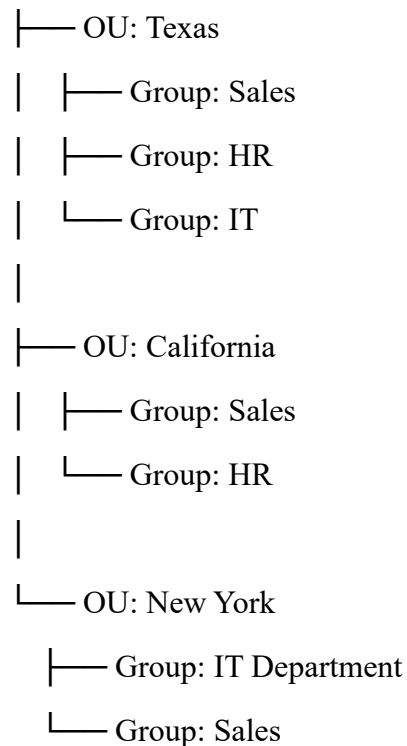
5. Domain Configuration

- **Domain Name:** Goldcorp.co
- **Server Name:** Goldcorp
- **Static IP:** 192.168.1.1
- **Roles Installed:**
 - Active Directory Domain Services (AD DS)
 - DNS Server

6. Organizational Units (OUs) and Groups

The directory structure was created as follows:

Goldcorp.co



7. Users and Group Memberships

Two test users were provisioned to demonstrate IAM principles:

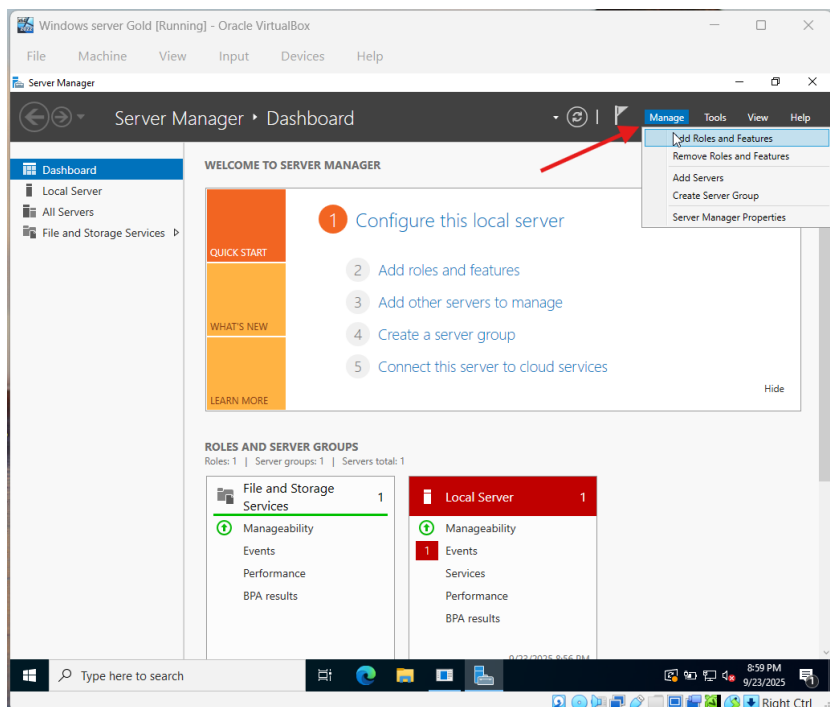
Username	OU	Group Membership	Assigned Policy
Michael.it.tx	Texas OU	IT	Disable removable disk access
Jane.sales.ny	New York OU	Sales	Unable to shut down

8. Implementation Steps

Step i: Virtual Environment Setup

- A Windows Server VM and two Windows 10 client VMs were launched in **VirtualBox**.
- All VMs were configured to use Bridged Networking, allowing them to have unique IP addresses and connect as standalone devices.

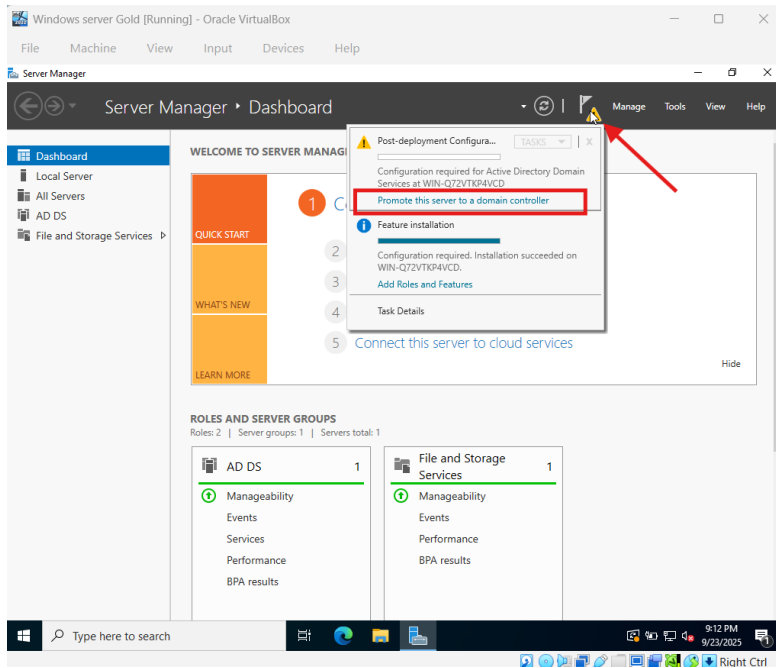
Step ii: Install Active Directory Domain Services (AD DS)



1. Open Server Manager → Manage → Add Roles and Features.

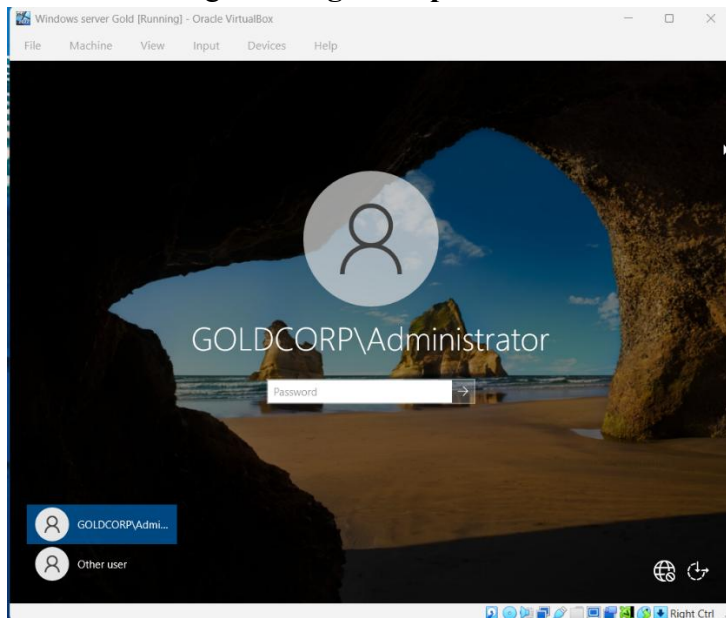
2. Select **Active Directory Domain Services (AD DS)**.
3. Follow the installation wizard until AD DS is successfully installed.
4. Verify that AD DS appears on the server dashboard.

Step iii: Promote Server to Domain Controller

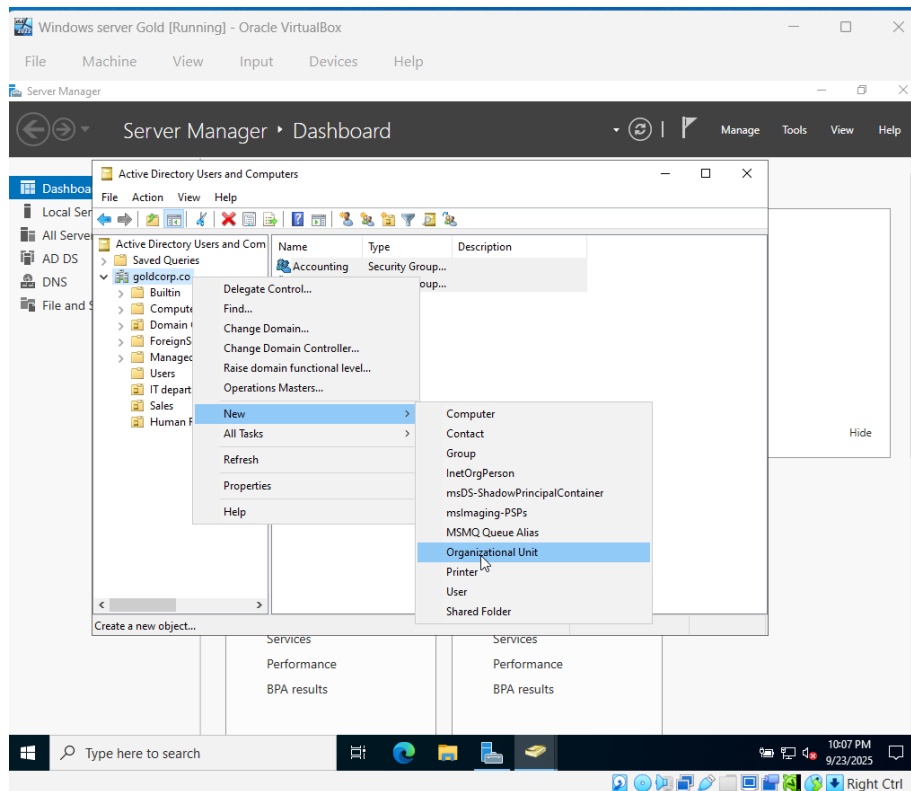


1. In **Server Manager**, click the yellow warning icon and select **Promote this server to a domain controller**.
2. Choose **Add a new forest** and set the root domain name as **goldcorp.co**.
3. Configure a Directory Services Restore Mode (DSRM) password.
4. Ignore the DNS delegation warning, proceed, and install.
5. The server restarts automatically, completing the promotion.

6. After reboot, log in with **goldcorp.co\Administrator** credentials.

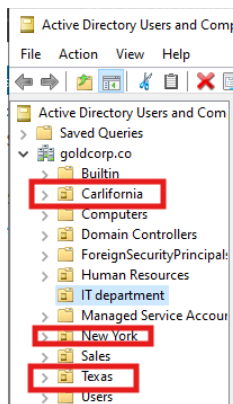


Step iv: Create Organizational Units (OUs)



1. Open **Active Directory Users and Computers (ADUC)**.
2. Right-click the domain **goldcorp.co**, select **New** → **Organizational Unit (OU)**.

3. Create three OUs: **Texas Branch, California Branch, New York Branch.**



Step v: Create Groups

1. Within each OU, create departmental groups: **Sales, IT, HR.**
2. These groups represent functional roles and allow role-based access management.

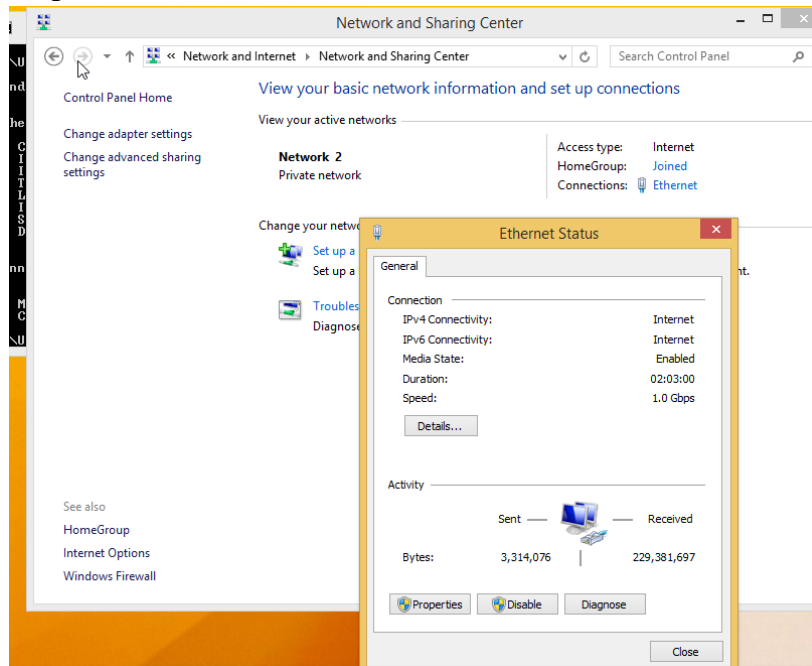
Step vi: Create User Accounts and Assign to Groups

1. Right-click the relevant OU/group → **New** → **User**.
2. Add users such as:
 - **Jane Doe** – New York Branch → Sales Department
 - **Michael Jackson** – Texas Branch → IT Department

3. Assign initial passwords and configure login credentials.
4. To assign a user to a group:
 - Right-click the user account and select **Properties**.
 - Navigate to the **Member Of** tab.
 - Click **Add** → **Advanced** → **Find Now**.
 - Select the appropriate group from the list (e.g., *Sales Department* or *IT Department*).
 - Click **Apply**, then **OK** to confirm.

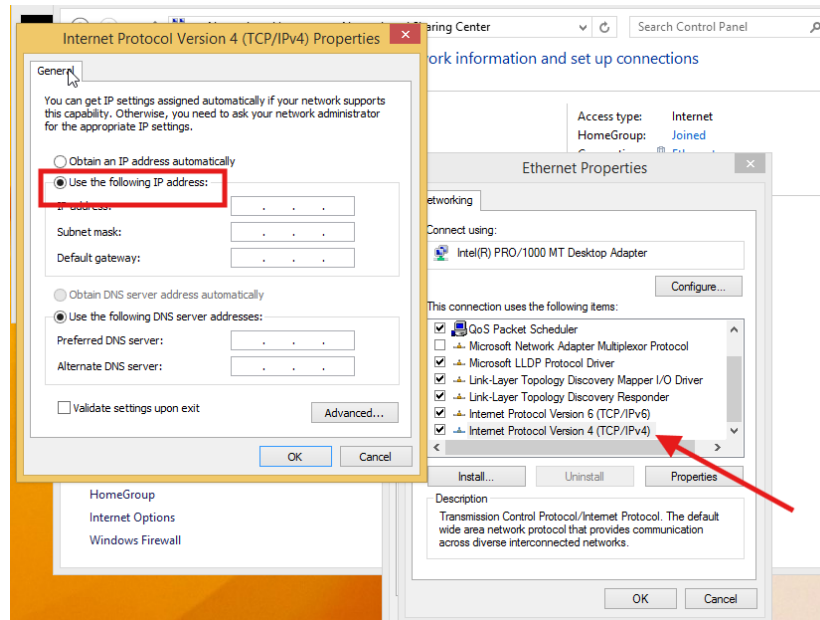
This ensures each user is linked to the correct departmental group, allowing group policies and permissions to be applied consistently.

Step vii: Join Client PCs to the Domain



1. On each client PC, configure network settings:
 - Navigate to **Control Panel** → **Network and Sharing Center** → **Ethernet Properties** → **internet protocol version 4 (ipv4)** → click use the following ip address.

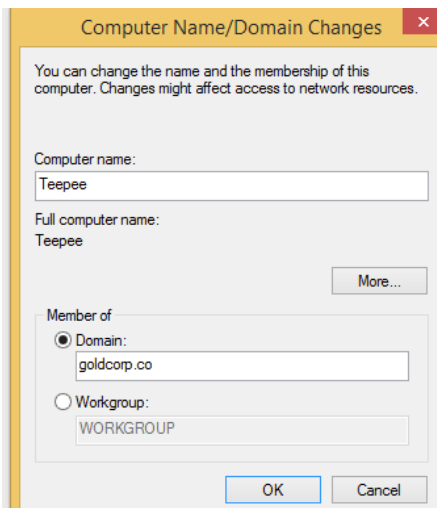
- Enter pc ip address and subnet, then set



the server IP address (192.168.1.66) as the **Preferred DNS Server**.

- Ensure the **default gateway** is 192.168.1.1 (the host router gateway).

2. Open System Properties → Computer Name/Domain Changes.



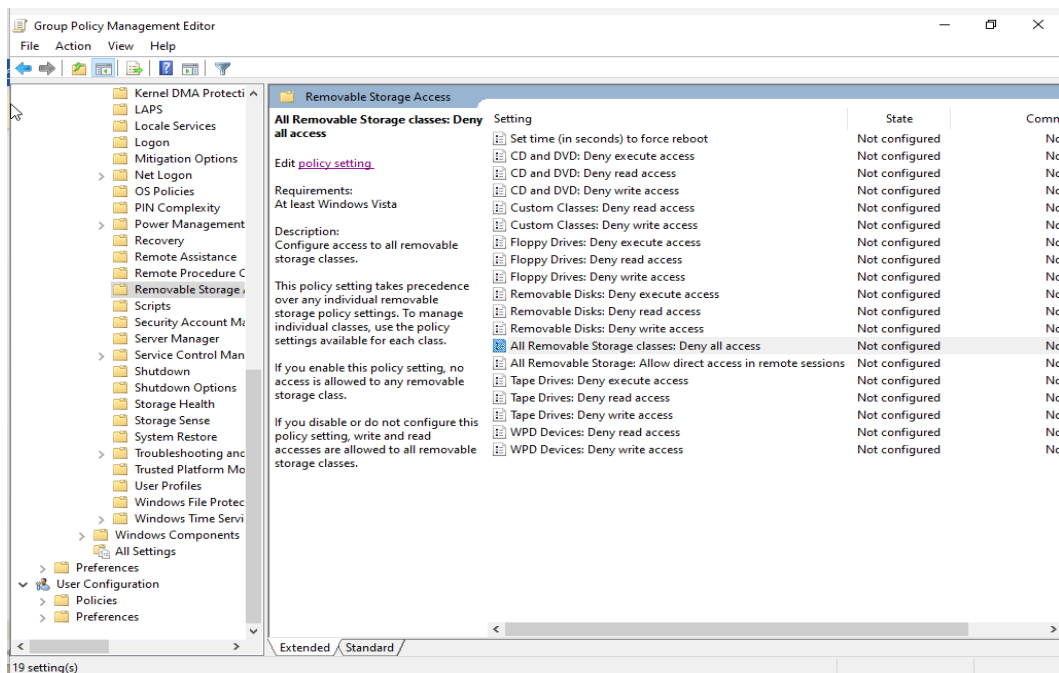
- Select **Domain** and enter **goldcorp.co**.
- Provide domain administrator credentials.
- Restart the PC to complete the domain join.

Step viii: Assign Group Policies (GPOs)

To enforce security restrictions, GPOs were created and linked to OUs.

Procedure:

- a. In **Server Manager** → **Tools**, open **Group Policy Management**.
- b. Expand the **Forest** → **Domains** → **goldcorp.co**.
- c. Right-click **Group Policy Objects** → **New**, name the GPO, and save.
- d. Right-click the new GPO → **Edit**.
- e. Configure policies as required:
 - **For Jane Doe (Sales Department):**
User Configuration → Administrative Templates → Start Menu and Taskbar → Remove and prevent access to the Shut Down, Restart, Sleep, and Hibernate commands.
 - **For John Doe (IT Department):**
Computer Configuration → Administrative Templates → System → Removable Storage Access → Deny all access.
- f. Link the GPO to the appropriate OU by right clicking the OU and selecting **Link an Existing GPO**.
- g. Force update policies on clients by going into the command prompt window, then type in the command: `gpupdate /force`



9. Testing and Validation

Two GPOs were created and linked to specific users:

- i. GPO Name: NoShutdown
 - Linked To: Texas OU (IT User – Michael)
 - Policy:
User Configuration → Administrative Templates → Start Menu and Taskbar → Remove and prevent access to the Shut Down, Restart, Sleep, and Hibernate commands
 - Result: Michael cannot shut down the assigned PC.
- ii. GPO Name: DisableRemovableDrives
 - Linked To: New York OU (Sales User – Jane)
 - Policy:
Computer Configuration → Administrative Templates → System → Removable Storage Access → Deny all access
Result: Jane cannot use USB or external drives.

10. Key Takeaways

- Successfully implemented a centralized IAM framework on Active Directory.
- Mapped business structure (regions and departments) into OUs and groups.
- Demonstrated access control enforcement using Group Policy Objects (GPOs).
- Learned how to provision and manage users, groups, and security policies.
- Applied identity governance principles in a real-world simulated enterprise environment.